Literature Review: Restaurant Service Robots

A review and comparison of three research papers supporting a restaurant service robot project

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1. Introduction

Robots in the restaurant sector have become an active area of research and application. This growth has been driven by labour shortages, the need to reduce physical contact after the COVID-19 pandemic, the falling cost of electronic components and, more recently, large language models (LLMs) that give robots far greater ability to understand and interact. A restaurant service robot must solve three problems at the same time:

1. **A technical problem:** moving safely and accurately through a tight, changing environment full of tables, chairs and moving people.
2. **An operational problem:** fitting into the restaurant's real workflow, from receiving the order, to passing it to the kitchen, to serving the food and finally handling payment.
3. **An interaction problem:** understanding changing and unexpected customer requests, and correcting its own mistakes during the task without burdening the customer.

This review examines three recent papers, each addressing one of these problems. The first focuses on autonomous navigation and control, the second on integrating the robot into restaurant operations, and the third on LLM-based task planning and customer interaction.

2. Selected Papers and Rationale

Item	Paper 1	Paper 2	Paper 3
Title	Autonomous navigation of a low-cost service robot in a dynamic restaurant environment using enhanced SLAM and GA-optimized fractional PI control	RoboServe: Design and Implementation of a Robotic Agent for Serving Food at Restaurants	Real-Time Service Robot Replanning via Simple Button Interaction for Improved Task Success and User Experience
Authors	Mac et al. (Hanoi University of Science and Technology)	Sayor et al. (Bangladesh / Australia / USA)	Terashima, Yano et al. (Kyushu Institute of Technology, Japan)
Venue	Results in Engineering, Elsevier, 2025	ICCA 2024 conference, ACM	arXiv:2609.17541, July 2026 (preprint)
Platform	Raspberry Pi 4 + Arduino + ROS	ESP32	Toyota HSR (mobile manipulator) + GPT-4o

Item	Paper 1	Paper 2	Paper 3
Navigation	Autonomous map-based SLAM navigation (RGB-D camera + IMU)	Black-line following with IR sensors	Not described in the paper
Focus	Perception, mapping, path planning, control	QR ordering, payment, voice interaction, delivery	LLM task planning, failure recovery, user experience

Reasons for Selection

- **Direct relevance:** all three papers target a restaurant service robot specifically, not a general mobile robot.
- **Recency:** published between 2024 and 2026; the third is among the most recent work in the field.
- **A range of technology levels:** from cheap line-following hardware (Paper 2), to map-based autonomous navigation (Paper 1), to a robot that plans its tasks with an LLM (Paper 3).
- **Complementarity:** each paper covers a layer the other two neglect: navigation, the service system, and intelligence and interaction.
- **Real implementation:** all three built and tested a real system; the third tested it with 224 users.

3. Strengths

Paper 1 (Mac et al., 2025)

- **Integrated navigation stack:** four modules combined on a single ROS platform:
  - Visual odometry (SURF + NNDR + PnP-RANSAC via RTAB-Map).
  - IMU and camera fusion with an Extended Kalman Filter (EKF) to reduce drift.
  - Hybrid path planning: A* for the global path and DWA for local avoidance of moving obstacles.
  - A fractional-order PI (FOPI) controller tuned by a Genetic Algorithm.
- **Measurable control results:** rise time reduced from 0.35 s to 0.17 s, and settling time from 0.58 s to 0.26 s.
- **Reusable details:**
  - DWA cost-function weights ($\alpha = 0.8$, $\beta = 0.1$, $\gamma = 0.1$).
  - The identified motor transfer function.
  - Genetic Algorithm parameters.
- **Statistical analysis:** each test was repeated 10 times and reported as mean ± standard deviation under different noise levels.
- **Free navigation:** no floor lines are needed, so the restaurant layout can change without modifying the infrastructure.

- **Honest about scope:** the authors acknowledge that the contribution is system integration rather than algorithmic novelty.

Paper 2 (RoboServe, 2024)

- **Complete service cycle:** table QR scan, then ordering website, then online payment (bKash), then database, then chef dashboard, then robot, then delivery, then return to the kitchen.
- **Practical smart features:**
  - Empty-tray detection with an IR sensor, so the robot returns automatically once the customer takes the food.
  - Voice alerts for people blocking the path, waiting 5 seconds before attempting to go around.
  - Multimodal control: touch and voice.
- **Very low cost:** about 5,600 BDT for the prototype, with an itemised cost table.
- **Clear, easy-to-implement algorithms:** pseudocode for going to a table, returning, line following, rotation and obstacle avoidance.
- **Per-feature functional evaluation** over 100 attempts:

Feature	Success rate
Line following	94%
Obstacle detection	95%
Delivery to the correct table	97%
Empty-tray detection	97%
Order taking	99%
Bill generation	98%
Data transmission	100%

- **Attention to management aspects:** covers coordination between the robot and staff and includes an economic analysis.

Paper 3 (Terashima et al., 2026)

- **LLM-based task planning:** the system sends the customer's order and the robot's skill list (move, grasp, place) to GPT-4o, which selects the next skill and its target step by step until the order is complete. The approach is based on SayCan and lets the robot cope with orders that change mid-task.
- **One-button failure recovery:** the customer presses a button on a tablet when the robot makes a mistake; the LLM infers what failed and replans. This handles failures the robot cannot detect on its own, such as mishearing an order or picking up the wrong drink.
- **Low user burden by design:** the customer does not need to explain the error, because failure analysis is delegated to the LLM.

- **Clear results with real users:**

Measure	Without button	With button
Number of users	91	133
Successful / total tasks	21 / 38	43 / 50
Success rate	55%	86%
"Was the robot's behaviour unexpected?" (out of 5, lower is better)	3.0	2.5
"Did you feel anxious about the robot's behaviour?" (out of 5, lower is better)	3.1	2.2

- **A notable finding:** the button was pressed in only 8 of 50 tasks, yet anxiety still fell. The authors suggest that simply having a way to correct the robot reassures customers.
- **A fully capable robot:** the HSR mobile manipulator takes orders by voice, grasps objects and serves them, with tasks designed to RoboCup@Home standards.
- **Honest about scope:** the authors acknowledge the success rate could be lower in a real restaurant where customers do not watch the robot constantly.

4. Weaknesses

Paper 1

- **Limited algorithmic novelty:** all components are off-the-shelf and used with default settings, such as RTAB-Map and YOLOv3 pre-trained on COCO without fine-tuning.
- **No perception evaluation:** precision, recall and F1 were not reported for obstacle detection.
- **Limited test environment:** a small restaurant of about 30 m^2, with no comparison against advanced planning or control methods.
- **Internal inconsistencies worth noting:**
  - The abstract says overshoot was "nearly eliminated", but Table 2 shows GA-FOPID overshoot (e.g. 0.57%) is higher than plain FOPID (0.13%) in most cases.
  - The GA population size is given as 30 in one place and 25 in another.
  - Robot dimensions are given as 50 × 40 × 1000 mm, an obvious unit error.
  - References [13] and [15] are duplicates, and the DOI for reference [2] is wrong.
  - A lidar (RPLidar-A1) appears in the experimental section without being mentioned in the system design.
- **Computational load unclear:** the paper does not explain how a Raspberry Pi 4 runs RTAB-Map and YOLOv3 together in real time, nor the frame rate achieved. Part of the processing appears to run on an external master station.
- **No service layer:** there is no ordering system, no real customer interaction and no mechanism to confirm food pickup.

- **Data not public:** available only "on request".

Paper 2

- **Primitive navigation:** relies entirely on a black floor line, two checkpoints and four fixed tables. Any change to the restaurant layout requires redrawing the lines.
- **Fragile obstacle avoidance:** the detour uses fixed timed moves (forward 2 seconds, then turn) with no map or distance sensing. This is unsafe in a crowded restaurant, even though the paper claims crowd navigation without testing it.
- **Operational limits:**
  - One order per trip.
  - Maximum payload of 1.5 kg.
  - Prototype body made of wood and paper.
  - No durability test was carried out.
- **Weak evaluation methodology:**
  - Success rates without a description of test conditions.
  - No measurement of time, speed or stopping accuracy.
  - No analysis of failure causes.
- **Optimistic economic analysis:** assumes the robot replaces two waiters without experimental evidence.
- **Self-serving comparison table:** the comparison criteria are chosen to match the paper's own features, mostly software features such as QR and payment, rather than performance metrics.
- **Writing and consistency issues:** for example, an ultrasonic sensor appears in the cost table while the text discusses IR sensors only.

Paper 3

- **Not peer-reviewed:** an arXiv preprint that has not yet gone through peer review.
- **Not a fully realistic setting:** a simulated restaurant at an event, with participants watching the robot closely the whole time.
- **Weak experimental comparison:**
  - Each condition was tested on a different group of users (91 vs. 133), not the same group.
  - No statistical test was applied to the differences.
  - Only 36 people answered the questionnaire in each condition.
- **Relies on customer attention:** if the customer does not notice a mistake, the button is not pressed; this did happen in the experiment, especially with children.
- **Missing details:** the paper does not describe the navigation method, the full skill list or the perception modules used.
- **Cost and speed:** the HSR is very expensive, and neither the cost nor the latency of GPT-4o is studied.

- **Inconsistent participant description:** the results section splits respondents into students and working adults, while the discussion refers to elementary-school children and adults.

5. Comparison of the Three Papers

The first table sets out the characteristics of the three papers side by side; the second gathers the main strengths and weaknesses of each.

Characteristics

Criterion	Paper 1 (Mac)	Paper 2 (RoboServe)	Paper 3 (Terashima)
Year	2025	2024	2026
Publication type	Peer-reviewed journal (Elsevier)	Peer-reviewed conference (ACM)	Non-peer-reviewed preprint (arXiv)
Cost	Low	Very low	Very high
Navigation	Advanced (SLAM + A* + DWA)	Primitive (line following)	Not described
Ordering and payment	None	Complete (QR + payment + chef dashboard)	Voice ordering, no payment
Intelligence and interaction	Weak	Moderate (voice and touch)	Advanced (LLM + failure recovery)
Error handling	None	Voice alert at obstacles	Replanning after button press
Testing with users	None	None	224 users + questionnaire
Evaluation rigour	Good (repeats + standard deviation)	Weak (rates without conditions)	Moderate (no statistical test)
Headline result	Settling time 0.58 → 0.26 s	Correct delivery 97%	Task success 55% → 86%

Strengths and Weaknesses

Paper	Main strengths	Main weaknesses
Paper 1 (Mac et al., 2025)	• Integrated autonomous navigation on ROS (SLAM + EKF + A* + DWA) • GA-tuned FOPI controller with a clear improvement in response • Statistical analysis with repeated tests • No floor lines required	• No algorithmic novelty; all components are off-the-shelf • No evaluation of obstacle-detection accuracy • Inconsistencies between abstract and tables • No ordering system or customer interaction
Paper 2 (RoboServe, 2024)	• Complete service cycle from order to payment	• Line-following navigation, inflexible • Timed-move obstacle avoidance, unsafe • One order per trip, 1.5 kg payload

Paper	Main strengths	Main weaknesses
	• Empty-tray detection for automatic return • Very low cost and easy-to-implement algorithms • Per-feature evaluation over 100 attempts	• Weak evaluation and optimistic economics
Paper 3 (Terashima et al., 2026)	• LLM-based task planning (GPT-4o) • One-button failure recovery during the task • Success rate raised from 55% to 86% • Tested with 224 users and a satisfaction survey	• Non-peer-reviewed preprint • Simulated restaurant; different users per condition • No statistical test; no navigation details • Expensive robot; relies on customer attention

What the Comparison Shows

- **Each paper is strong in one layer:** the first in motion, the second in service, the third in intelligence and interaction.
- **A clear cost–capability trade-off:** the more intelligence, the higher the cost. The challenge is delivering Paper 3's intelligence on hardware close to that of Papers 1 and 2.
- **No paper combines all three layers:** autonomous navigation, a complete service system and intelligent planning with failure recovery, in one low-cost robot.

6. How These Papers Support My Project

6.1 Choosing the Right Navigation Level for the Budget

Papers 1 and 2 represent the two ends of the navigation spectrum:

- **Limited budget and time:** starting with the RoboServe approach (ESP32 + IR sensors + floor line) gives a working prototype quickly.
- **A flexible robot closer to real use:** following Paper 1 (ROS + RGB-D camera or lidar + SLAM).
- **A realistic middle ground:** start with line following, while designing the project so it can later be upgraded to map-based navigation.

6.2 Reusable Components

From Paper 1:

- RTAB-Map for mapping and localisation.
- The ROS navigation stack with A* and DWA, using the cost-function weights as a tuning starting point.
- IMU–camera fusion via EKF, which the ROS robot_localization package implements directly.

- The method of identifying the motor model (transfer function), then tuning the controller with a Genetic Algorithm using the ITAE criterion.

From Paper 2:

- The ordering architecture: QR, then website (PHP/MySQL or a modern alternative), then kitchen dashboard, then robot command.
- Empty-tray detection as the return trigger, a very simple and effective idea.
- The alert, wait, then detour logic when a person blocks the path.

From Paper 3:

- A "something went wrong" button on the robot's screen, which can be linked to an LLM through an API without needing an expensive robot.
- Breaking the robot's work into simple skills (go to table, wait for pickup, return to kitchen) that the LLM chooses between according to the order.
- Showing the robot's state clearly on screen, so customers know when to intervene without constant monitoring.

6.3 Testing and Evaluation Methodology

- Test each function over many attempts and record the success rate, as in Paper 2.
- Repeat control tests and report mean ± standard deviation, as in Paper 1.
- Run a five-point satisfaction survey with real customers, as in Paper 3.
- Add what all three papers lack:
  - Delivery time and stopping accuracy at the table.
  - Obstacle detection accuracy (precision/recall).
  - Performance with moving people present.
  - A statistical test when comparing two conditions, ideally with the same group of users.

6.4 A Research Gap the Project Can Fill

None of the three papers combines **free autonomous navigation**, a **complete service system** (ordering, payment, delivery confirmation) and **intelligent planning with failure recovery** on low-cost hardware. A project that integrates these layers can be presented as a clear contribution, especially with:

- A more recent obstacle detector than YOLOv3 (such as newer YOLO versions), with measured accuracy.
- Testing in a real restaurant, or with real disturbance from moving people.
- Support for more than one order per trip.

6.5 Mistakes to Avoid When Writing the Report

The three papers show common mistakes that should be avoided:

- Claims in the abstract that do not match the tables.

- Contradictory values across sections.
- Comparisons using criteria chosen to favour one's own work.
- Cost analysis not backed by experimental evidence.
- Comparing two different user groups without a statistical test.

7. Conclusion

The first paper provides a solid technical foundation for autonomous navigation and precise velocity control; the second offers a practical model for integrating the robot into the full restaurant service cycle; and the third adds an intelligence layer that lets the robot understand changing orders and correct its mistakes with the customer's help. Combining the strengths of all three, while addressing their weaknesses in evaluation, testing and cost, provides a clear framework for building an intelligent, low-cost and deployable restaurant service robot.

References

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